

Econ 590: News Driven Business Cycles

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0. Introduction

- ▶ These slides build on a survey we wrote “*News Driven Business Cycles: Insights and Challenges*”, Journal of Economic Literature, 2015.

0. Introduction

The news view of business cycles: General idea

- ▶ Boom/bust may be the result of agents having incentives to continuously anticipate the economy's future demands.
- ▶ Properly anticipating a future need $\rightsquigarrow$, gains by trying to preempt the market and invest early as to make goods readily available when the predicted needs eventually appear.
- ▶ If many agents adopt similar behavior, because they receive related news about future developments, $\rightsquigarrow$ a boom period.

0. Introduction

The news view of business cycles: General idea

- ▶ However, by the very fact that such behavior involves speculation it will be subject to error.
- ▶ In the cases of error, the economy will have over-invested as the anticipated demand will not materialize.
- ▶ This will cause a recession and a process of liquidation.
- ▶ Both the boom and the bust are direct consequences of people's incentive to speculate on information related to future developments of the economy.

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An example: the satellite industry

- ▶ 1990s: anticipation of high demand for bandwidth (related to the internet boom)
- ▶ The launching of telecommunication satellites exploded.
- ▶ In 1998, the satellite industry launched 150 satellites, 300 percent growth compared to 1993.
- ▶ The boom was not caused by a technological improvement nor an increase of current demand.
- ▶ The growth rate for demand for satellite bandwidth has been 31% between 1995 and 2003, while the supply of satellite bandwidth grew by 54% during this same timeframe
- ▶ At the beginning of the new millenium, demand failed to materialize $\rightsquigarrow$ *transponder glut*.
- ▶ The number of satellites launched went down to 75 in 2001 and 69 in 2003, against 150 in 1998.

0. Introduction

An example: the satellite industry (*continued*)

- ▶ Variety, March 25, 1998 : *“**Sat glut in Arab skies:** With a glut of satellite transponder space available in the crowded Mideastern skies, the region looks set for a **cutthroat price war**. Arabic channels already have the multinational Arabsat as well as Eutelsat and Intelsat to choose from, and in late April these will be joined by Egypt’s Nilesat, which will have **72 transponders available for rental**.”*

0. Introduction

An example: the satellite industry (*continued*)

- ▶ Satellite News, Nov 4, 2002 : “Companies that purchased satellite transponder capacity during the *industry's halcyon days* are now *struggling to resell the capability* in the face of falling market prices.
One example is Globecom Systems [GCOM], a Hauppauge, N.Y.-based provider of end-to-end satellite-based communications services. To rid itself of carrying *\$400,000 in unused capacity each month*, Globecom reached a broad-based agreement with one of its vendors to cut the company's future obligations for satellite bandwidth capacity.”

0. Introduction

News?

- ▶ Let's consider an exogenous variable, say productivity θ
- ▶ $Y = \theta F(K, H)$
- ▶ $\theta_t = \rho\theta_{t-1} + \varepsilon_t$
- ▶ $\theta_t = \rho\theta_{t-1} + \varepsilon_{t-q}$
- ▶ Signal $S_t = \varepsilon_t + \eta_t$, where η_t is a Gaussian white noise error term.
- ▶ **Note :** This is an example, News need not to be only about technology.

0. Introduction

Roadmap

1. Present some suggestive informal evidence
2. Discuss of a model in which expectation shocks matter in a relevant way
3. Discuss some extensions

0. Introduction

- ▶ **Note 1:** No literature review in this talk (and not even references to existing work)
- ▶ **Note 2:** No discussion of structural evidence (VARs, estimated models)

0. Introduction

Roadmap

1. Informal Evidence
2. An Analytical Framework
3. Discussion / Extensions

1. Informal Evidence

- ▶ Here I display some informal evidence showing that there is evidence of news driven BC and liquidation cycles
- ▶ By liquidation cycle I mean the following sequence
 - × First phase of capital accumulation caused by rosy expectations.
 - × Second phase of downwards expectations revision.

1. Informal Evidence

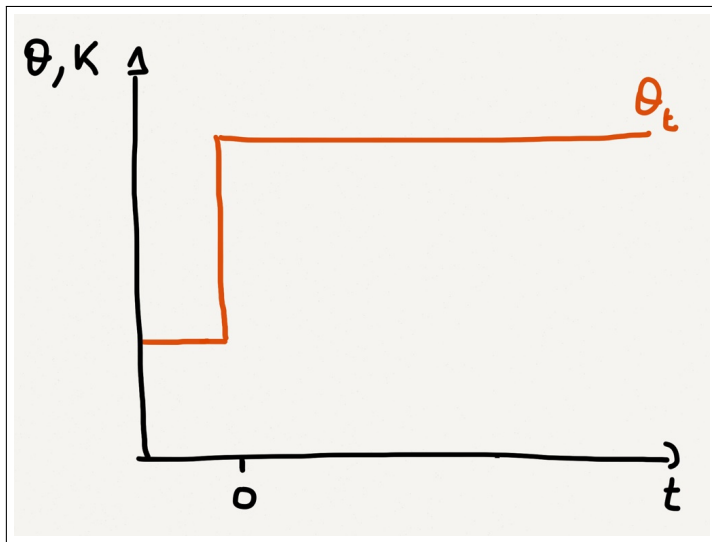
- ▶ Think of some stochastic G.E. model in which shocks to TFP are random walk.

$$\theta_t = \theta_{t-1} + \epsilon_t$$

- ▶ The state of the economy is summarized by $\frac{K}{\theta}$

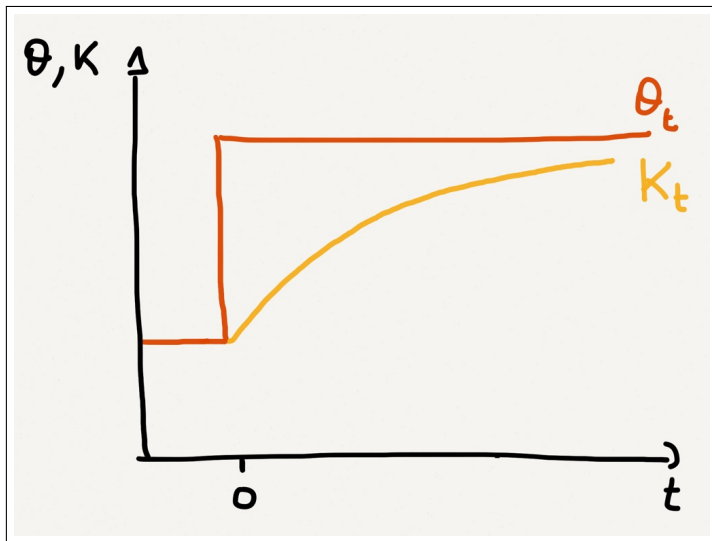
1. Informal Evidence

Figure 1: A surprising increase in θ



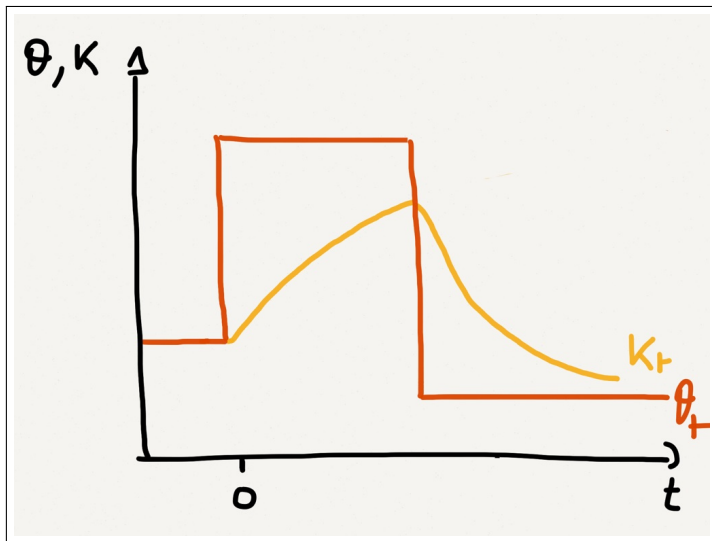
1. Informal Evidence

Figure 2: A surprising increase in θ



1. Informal Evidence

Figure 3: A surprising decrease in θ



1. Informal Evidence

Predictions for $\frac{K}{\theta}$

- ▶ If $\frac{K}{\theta}$ is low, the model predicts that employment should be high
- ▶ If $\frac{K}{\theta}$ is high, then employment should be low as there are low returns to capital accumulation + wealth effect.
- ▶ According to a “surprise” view, employment and $\frac{K}{\theta}$ should be strongly negatively correlated.
- ▶ That predictions holds in the same way if nominal rigidities are added (although it depends on the monetary rule)

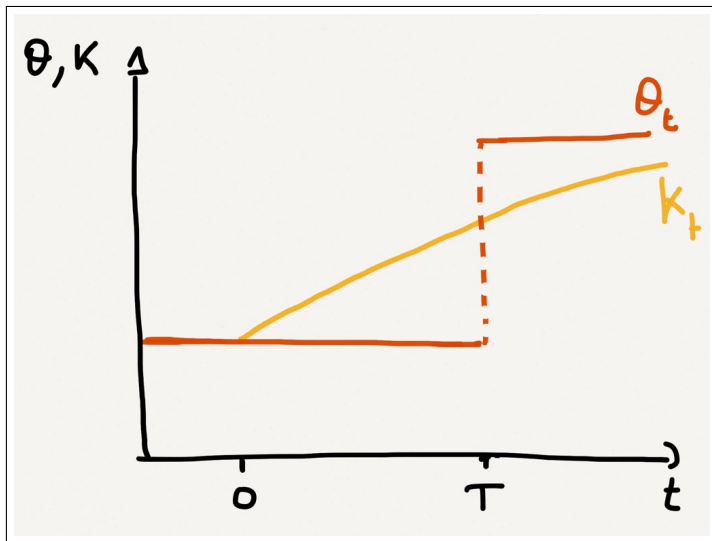
1. Informal Evidence

The News view

- ▶ The news view of business cycles suggest that employment and $\frac{K}{\theta}$ should have a modest positive correlation

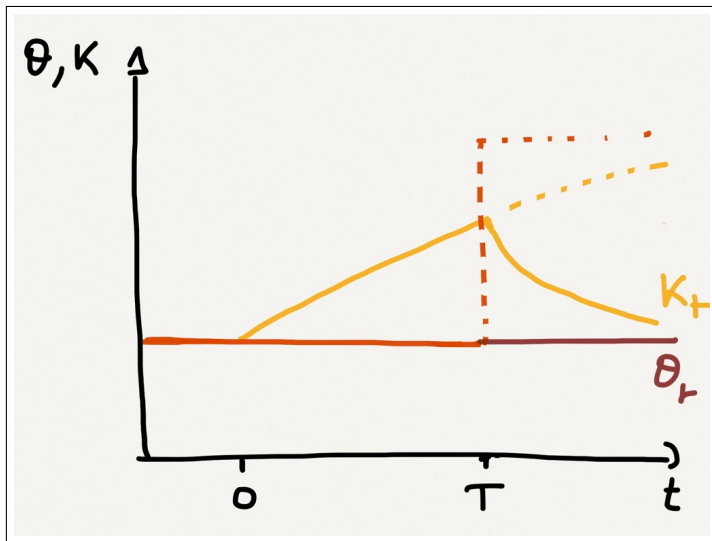
1. Informal Evidence

Figure 4: A news about future increase in θ



1. Informal Evidence

Figure 5: A downward revision of θ



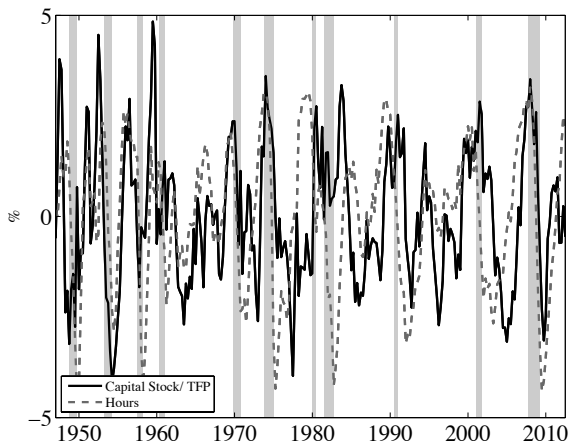
1. Informal Evidence

Predictions for $\frac{K}{\theta}$

- ▶ If employment booms arise because capital accumulation anticipates growth in TFP, then employment and $\frac{K}{\theta}$ should be positively correlated in booms.
- ▶ Major recessions in the news view of business cycles arise when $\frac{K}{\theta}$ is high and expectations no longer support such a high capital stock and this leads to a recession
- ▶ The second force should contribute to a negative correlation between employment and $\frac{K}{\theta}$
- ▶ Since on average agents should be right more often than wrong, it suggests that employment and $\frac{K}{\theta}$ should be positively correlated according to the news view.

1. Informal Evidence

Figure 6: $\frac{K}{\theta}$ and H , US 1948-2012



► correlation = .30

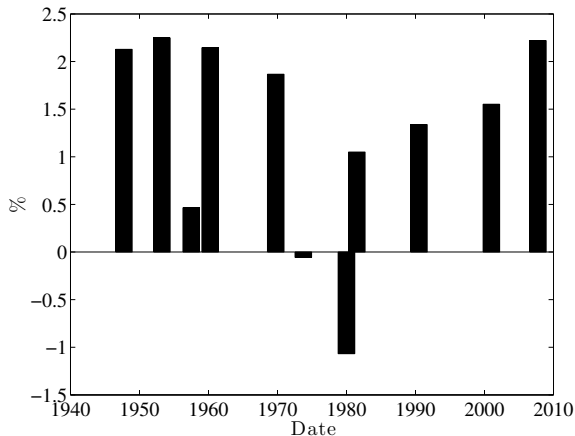
1. Informal Evidence

Recession episodes

- ▶ In the “surprise” view, recessions are caused by negative technological shock.
- ▶ Major recessions are therefore caused by major shocks to TFP, that are not correlated with the level of $\frac{K}{\theta}$.
- ▶ In the news view, major recessions occur when $\frac{K}{\theta}$ is high and expectations revised.
- ▶ One should expect $\frac{K}{\theta}$ to be *high* the periods before a major recession.

1. Informal Evidence

Figure 7: Mean HP deviation of $\frac{K}{\theta}$ the year of a NBER peak



- $\frac{K}{\theta}$ is high the year before a recession starts.

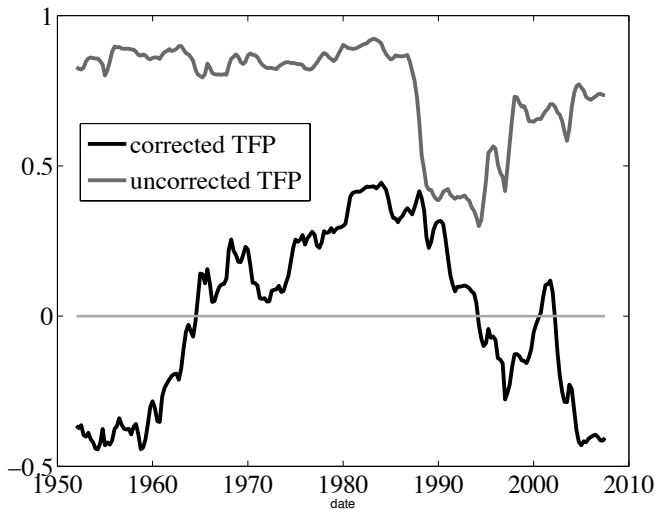
1. Informal Evidence

Shocks

- ▶ Why are News/Revision shocks not only possible but also of interest?
- ▶ Because technological surprises are not the story (and no obvious observed “demand shocks”)

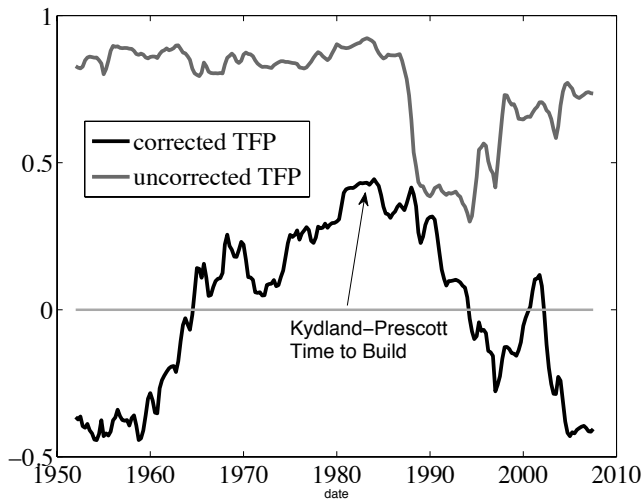
1. Informal Evidence

Typical TFP shocks are not good candidates for the post Volcker period



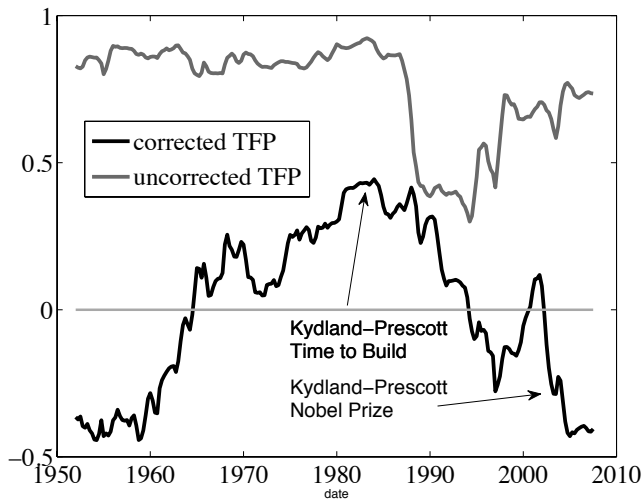
1. Informal Evidence

Typical TFP shocks are not good candidates for the post Volcker period



1. Informal Evidence

Typical TFP shocks are not good candidates for the post Volcker period



1. Informal Evidence

Neither are IST shocks

- ▶ If IST shocks, investment is cheap in booms, expensive in recessions.
- ▶ Various measures of the relative price of investment, deflating with core CPI, correlations with Hours:

Variable	1960Q1-2012Q3	Post-Volcker
Fixed I	0.42	0.76
Struct.I	0.44	0.75
Equip.I	-0.25	0.17
Resid.I	0.70	0.80
SP500	0.31	0.56

0. Introduction

Roadmap

1. Informal Evidence
2. [An Analytical Framework](#)
3. Discussion / Extensions

2. An Analytical Framework

Overview

- ▶ A quite simple analytical model
- ▶ Where investment is explicitly driven by future profit opportunities
- ▶ It shows how boom/bust cycles can be generated
- ▶ The discipline for some assumptions and the choice of functional forms: tractability, not quantitative relevance.

2. An Analytical Framework

Households



$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\ln(C_t) + \nu \cdot (\bar{L} - L_t^T) \right]$$

► The household can:

- × buy some or all of its consumption on the market
- × supply labor time to the market
- × produce household consumption
- × buy one-period bonds
- × trade in firm shares.



$$C_t^M + B_{t+1} + \sum_{i=1}^N P_{it} Z_{it+1} = w_t L_t^M + B_t(1+r_t) + \sum_{i=1}^N P_{it} Z_{it} + \sum_{i=1}^N d_{it} Z_{it+1}$$

- Home production: $C_t^H = \alpha L_t^H$
- $C_t = C_t^M + C_t^H$ and $L_t^T = L_t + L_t^H$

2. An Analytical Framework

Production side

- ▶ Set of intermediate sectors $i = 1, \dots, N$
- ▶ final good sector that aggregates the intermediate goods into a market consumption good according to:

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^\phi \right]^{\frac{1}{\phi}}, \quad \phi < 1$$

- ▶ Intermediate sectors and final market consumption good sector are competitive.
- ▶ Market consumption good is the numéraire
- ▶ Price of the intermediate goods: P_{it}

2. An Analytical Framework

Intermediate goods (1)

- ▶ The representative firm in sector i produces the intermediate good X_{it} :

$$X_{it} = K_{it}^{\gamma} (L_{it}^P)^{1-\gamma}, \quad 0 < \gamma \leq 1$$

- ▶ The firm also hires labor L_{it}^I to build up its capital stock:

$$K_{it+1} = I_{it} + (1 - \delta)K_{it},$$

with $I_{it} = \ln(L_{it}^I)$

- ▶ Let dividends be $d_{it} = P_{it}X_{it} - w_t L_{it}$

2. An Analytical Framework

Intermediate goods (2)

- The firm maximizes expected discounted stream of dividends:

$$\max_{L_t^P, L_t^I} E_0 \sum_{t=0}^{\infty} \frac{\beta^t C_0}{C_t} (P_{it} X_{it} - w_t L_{it})$$

subject to

$$X_{it} = K_{it}^{\gamma} (L_{it}^P)^{1-\gamma}, \quad 0 < \gamma \leq 1$$

$$K_{it+1} = \ln(L_{it}^I) + (1 - \delta) K_{it}$$

- Firms are assumed to finance themselves by retained earnings
- Shares in each sector are normalized to 1.

2. An Analytical Framework

Walrasian equilibrium

Result 1

The equilibrium is characterized by a constant wage rate $w_t = \alpha$ and a constant interest rate $(1 + r_t) = \frac{1}{\beta}$.

2. An Analytical Framework

Firm Behavior (1)

- ▶ This result makes the problem of the intermediate firm easier to solve.
- ▶ Optimal investment:

$$I_{it} = \ln \left(\frac{\beta}{\alpha} \gamma (1 - \gamma)^{\frac{1-\gamma}{\gamma}} E_t \sum_{j=0}^{\infty} (\beta(1 - \delta))^j P_{it+j+1}^{\frac{1}{\gamma}} \right)$$

- ▶ Investment is purely forward looking
- ▶ Need to build capacity in anticipation of the economy's future needs.

2. An Analytical Framework

Firm Behavior (2)

- Optimal labor demand:

$$\begin{aligned}L_{it} &= L_{it}^P + L_{it}^I \\&= K_{it} \left(\frac{(1-\gamma)P_{it}}{\alpha} \right)^{\frac{1}{\gamma}} \\&\quad + \frac{\beta}{\alpha} \gamma (1-\gamma)^{\frac{1-\gamma}{\gamma}} E_t \sum_{j=0}^{\infty} (\beta(1-\delta))^j P_{it+j+1}^{\frac{1}{\gamma}}\end{aligned}$$

- Employment itself is partially determined by expectations of future capital prices, that signal future (expected) needs of capital.

2. An Analytical Framework

Firm Behavior (3)

- ▶ Intermediate firms need to form expectations about the future price of their output P_{it+j} :

$$P_{it+j} = \frac{1}{N} \left(\frac{C_{t+j}^M}{X_{it+j}} \right)^{1-\phi} \theta_{it+j}^{\phi}$$

- ▶ To form expectations of future prices, firms will want to have information relevant for predicting aggregate consumption, aggregate production in their sector and technological change in their sector.
- ▶ Some of this information may be contained in current news (for example signals about future changes in the θ 's).
- ▶ This is in general a hard problem to solve (see later), but tractable in the case $\phi = 0$.

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^{\phi} \right]^{\frac{1}{\phi}}$$

2. An Analytical Framework

The case $\phi = 0$

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^\phi \right]^{\frac{1}{\phi}}$$

- ▶ When $\phi = 0$, P_{kt+j} writes

$$P_{kt+j} = \frac{1}{N} \frac{C_{t+j}^M}{X_{kt+j}} = \frac{1}{N} \frac{\left(\prod_{i=1}^N (X_{it+j})^{\frac{1}{N}} \right)}{X_{kt+j}} \left(\prod_{i=1}^N (\theta_{it+j})^{\frac{1}{N}} \right)$$

- ▶ The firm needs predict future market consumption and the future output level that will be offered in his sector (its market share, the demand for its own good).
- ▶ No need to care only about predicting technology levels.
- ▶ The News view should not be taken literally as RBC with a twist
- ▶ However...

2. An Analytical Framework

Symmetric equilibrium

- ▶ However ...
- ▶ given that this price function is symmetric for all sectors,
- ▶ a symmetric equilibrium implies that the output prices take the following simple form for all sectors:

$$P_{kt+j} = \frac{1}{N} \left(\prod_{i=1}^N (\theta_{it+j})^{\frac{1}{N}} \right) \equiv \Theta_{t+j}$$

- ▶ Back to the RBC view of news shocks (but very specific as $\phi = 0$ is needed)

2. An Analytical Framework

News and noise

- ▶ Technology index Θ follows an autoregressive process:

$$\Theta_t = \bar{\Theta} + \rho\Theta_{t-1} + \epsilon_{t-q}, \quad 0 \leq \rho < 1$$

- ▶ ϵ_t is a Gaussian white noise process.
- ▶ Signal

$$S_t = \epsilon_t + \eta_t$$

- ▶ η_t is a Gaussian white noise error term.
- ▶ Let's study the response of the economy to news and noise in that tractable case where $\phi = 0$ and $\gamma = 1$, so that L is fully forward looking.

$$X_{it} = K_{it}^{\gamma} (L_{it}^P)^{1-\gamma}$$

2. An Analytical Framework

When $\gamma = 1$ (1)

- Labour demand writes:

$$L_{it} = \frac{\beta}{\alpha} E_t \sum_{j=0}^{\infty} (\beta(1 - \delta))^j P_{it+j+1}$$

- Let $\psi = \frac{\sigma_{\epsilon}^2}{\sigma_{\eta}^2 + \sigma_{\epsilon}^2}$ captures the information content of the signal S_t with respect to ϵ_t

2. An Analytical Framework

When $\gamma = 1$ (2)

- Relative deviation of L from SS:

$$\begin{aligned}\hat{L}_t &= \frac{(1 - (1 - \delta)\beta)}{(1 - \rho(1 - \delta)\beta)} \times \\ &\quad \left[\sum_{j=0}^{q-1} ((1 - \delta)\beta)^{q-j-1} \psi \epsilon_{t-j} + \sum_{j=q}^{\infty} \rho^{j-q+1} \epsilon_{t-j} \right] \\ &\quad + \left[\sum_{j=0}^{q-1} ((1 - \delta)\beta)^{q-j-1} \psi \eta_{t-j} \right]\end{aligned}$$

- In comparison

$$\hat{\Theta}_t = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-q-j}$$

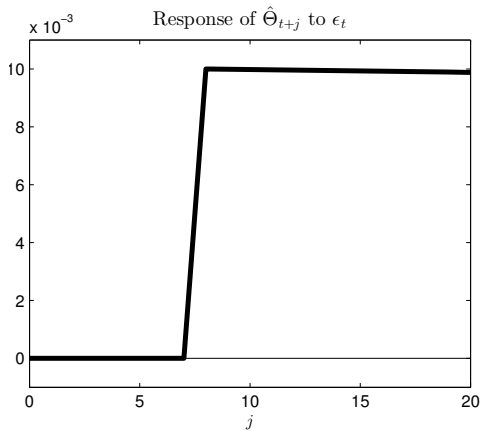
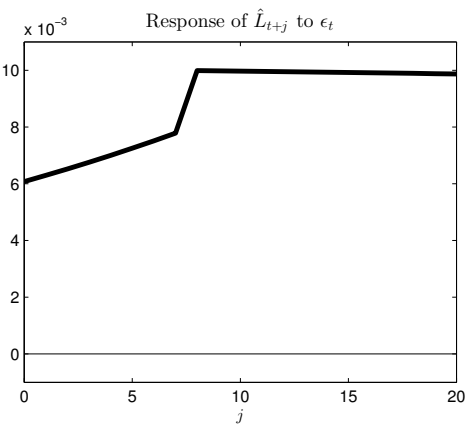
2. An Analytical Framework

IRF (1)

- ▶ Numerical illustration: $\beta = .99$, $\delta = .025$, $\rho = .999$, $q = 8$, $\psi = .8$, $\alpha = 25$ and $\kappa = 0.1$.
- ▶ Consider successively news and noise shocks.

2. An Analytical Framework

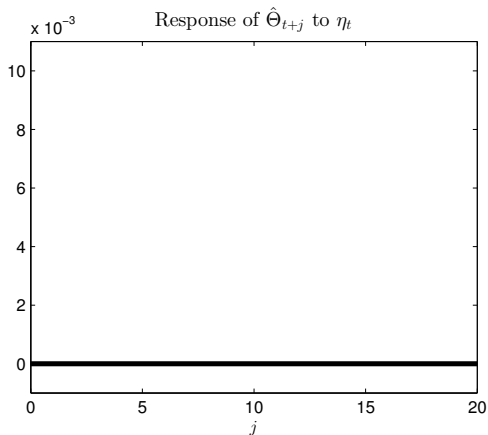
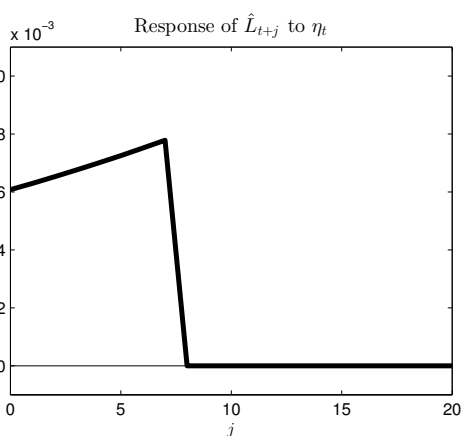
IRF to news



- On receiving the news employment starts to increase immediately, while $\hat{\Theta}$ does not change

2. An Analytical Framework

IRF to noise



- A key feature of this model is that it creates potentially large reversals in employment as the result of agents re-evaluating their information.

2. An Analytical Framework

Asset prices

- Define

$$\mathcal{P}_{it} = E_t \sum_{j=0}^{\infty} \beta^j d_{it+j} = E_t \sum_{j=0}^{\infty} \beta^j (P_{it+j} K_{it+j} - w_t L_{it+j})$$

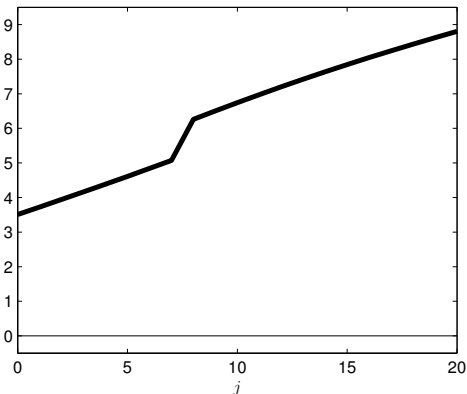
- First-order approximation:

$$\begin{aligned} \widehat{\mathcal{P}}_t &= \sum_{k=0}^{q-1} \left\{ \tilde{K} \frac{\beta^{q-k}}{1-\rho\beta} \psi + [1 + \alpha(1-\delta)\tilde{L}] B_{k-1} \psi \right\} \epsilon_{t-k} \\ &+ \sum_{k=q}^{\infty} \left\{ \tilde{K} \frac{\rho^{k-q}}{1-\rho\beta} + [1 + \alpha(1-\delta)\tilde{L}] [A_{k-1} + B_{k-1}(1-\delta)^{k-q} \psi] \right\} \epsilon_{t-k} \\ &+ \sum_{k=0}^{q-1} \left\{ \tilde{K} \frac{\beta^{q-k}}{1-\rho\beta} \psi + [1 + \alpha(1-\delta)\tilde{L}] B_{k-1} \psi \right\} \eta_{t-k} \\ &+ \sum_{k=q}^{\infty} \left\{ [1 + \alpha(1-\delta)\tilde{L}] B_{q-1} (1-\delta)^{k-q} \psi \right\} \eta_{t-k} \end{aligned}$$

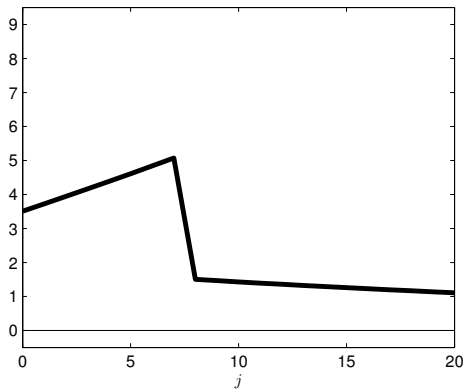
2. An Analytical Framework

Stock prices

Response of $\hat{P}_{t+j}^s$ to ϵ_t



Response of $\hat{P}_{t+j}^s$ to η_t



- **Remark:** If L bears adjustment costs, the stock price would be the best indicator of the agents' information and would be the first variable to move in response to news. (key insight for identification of news)

2. An Analytical Framework

Market consumption

- ▶ Market consumption is given by:

$$C_t^M = \Theta_t \sum_{j=0}^{\infty} \ln(1 - \delta)^j \frac{L_{t-j-1}}{N}$$

- ▶ Does not move on impact, then increases for q periods, then either collapse (noise) or increase further (news).
- ▶ Hard to make it increase on impact in “standard” models

0. Introduction

Roadmap

1. Informal Evidence
2. An Analytical Framework
3. Discussion / Extensions

3. Discussion / Extensions

Alternative news formulation (1)

- ▶ Instead of $\Theta_t = \bar{\Theta} + \rho\Theta_{t-1} + \epsilon_{t-q}$ plus a signal,
- ▶ Assume

$$\Theta_t = \bar{\Theta} + \rho\Theta_{t-1} + \epsilon_{1t-q} + \epsilon_{2t}$$

- ▶ There is no noise but a news and a surprise shock
- ▶ Response of L is then (for $\rho = 0$)

$$\begin{aligned}\hat{L}_t &= \frac{(1 - (1 - \delta)\beta)}{(1 - \rho(1 - \delta)\beta)} \times \\ &\quad \left[\sum_{j=0}^{q-1} ((1 - \delta)\beta)^{q-j-1} \epsilon_{1t-j} + \sum_{j=q}^{\infty} \rho^{j-q+1} \epsilon_{1t-j} \right] \\ &\quad + \left[\sum_{j=0}^{\infty} \rho^{j+1} \epsilon_{2t-j} \right]\end{aligned}$$

3. Discussion / Extensions

Alternative news formulation (2)

- Recall

$$\Theta_t = \bar{\Theta} + \rho\Theta_{t-1} + \epsilon_{1t-q} + \epsilon_{2t}$$

- Response to the news is similar to the previous one, except that it is not downplayed by the factor ψ .
- The surprise shock creates a standard response
- This is the type of information structure that is implicit in the VAR literature on the news shocks identification.
- The alternative structure with noise needs more structure to be recovered from the data (structural models needed)

3. Discussion / Extensions

The non-trivial information processing problem, and the interpretation of errors (1)

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^{\phi} \right]^{\frac{1}{\phi}}$$

- ▶ With $\phi = 0$, the model boils down to a boring expectation of an exogenous variable Θ .
- ▶ When $\phi \neq 0$, the prediction problem faced by firms becomes much more involved.
- ▶ To see this, let's consider another specific version of the model:
 - × $\phi \neq 0$
 - × $\delta = 1$
 - × $\theta_{jt} = \bar{\theta} + \epsilon_{jt-1}$ (*iid*)
 - × $S_{it} = \epsilon_{jt} + \eta_{it}$ (*one period news*)

3. Discussion / Extensions

The non-trivial information processing problem, and the interpretation of errors (2)

- In such a case

$$L_{jt} = E_{jt} \left[\frac{\beta}{\alpha} P_{jt+1} \right]$$

- and

$$E_{jt} P_{jt+1} = E_{jt} \left[\frac{\sum_{i=1}^N (\epsilon_{it+1} X_{it+1})^\phi}{X_{jt}} \right]^{1-\sigma} \epsilon_{jt+1}^\phi$$

- with

$$X_{jt} = \ln(L_{jt})$$

- The firm j information set is the collection $\{S_{it}\}_{i=1}^N$ (or a subset of it)

$$E_{jt} P_{jt+1} = E_{jt} \left[\frac{\sum_{i=1}^N (\epsilon_{it+1} \ln \left(E_{it} \left[\frac{\beta}{\alpha} P_{it+1} \right] \right)^\phi}{X_{jt}} \right]^{1-\sigma} \epsilon_{jt+1}^\phi$$

3. Discussion / Extensions

The non-trivial information processing problem, and the interpretation of errors (3)

► Recall

$$E_{jt}P_{jt+1} = E_{jt} \left[\frac{\sum_{i=1}^N (\epsilon_{it+1} \ln(E_{it}[\frac{\beta}{\alpha} P_{it+1}]))^\phi}{X_{jt}} \right]^{1-\sigma} \epsilon_{jt+1}^\phi$$

- Non linear problem
- Complementarity structure: firms in one sector will want to increase their production if they expect others to increase their production, regardless of the actual news received.
- There is a rational expectation solution with common information
- Errors by some firms lead to errors by other firms: if limited ability to process information, cycles can occur.
- News shocks generally coined as “augmented TFP” shocks, but this is a restricted view

3. Discussion / Extensions

Recessions as Liquidation Periods (1)

- ▶ Booms are periods in which expectations might be rosy (because of the noise).
- ▶ The post-bust periods (after expectation revisions) are periods of *liquidation*, as investment falls and capital is depleted.
- ▶ However, I and L simply return to SS because we have a simplified setup.
- ▶ Model modification:
 - × $C_t^M = \Theta_t X_t - \kappa \frac{X_t^2}{2}$ (DRS)
 - × Keep noisy signals on Θ
 - × $I_{it} = 2(L_{it}^I)^{\frac{1}{2}}$

3. Discussion / Extensions

Recessions as Liquidation Periods (2)

- Solution is

$$K_{t+1} = \lambda K_t + \frac{\beta\lambda}{1-\delta} \sum_{j=0}^{\infty} (\beta\lambda)^j E_t \Theta_{t+1+j}$$

- and

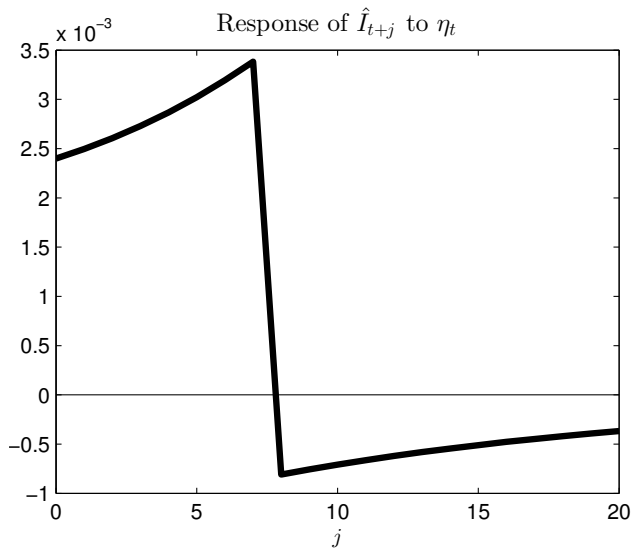
$$\hat{l}_t = \Omega \sum_{j=0}^{\infty} \phi_j \eta_{t-j}$$

where

$$\Omega \equiv \frac{\beta (\beta\lambda)^{q+1}}{(1-\delta)(1-\rho\beta\lambda)(1-\beta\lambda^2)}$$
$$\phi_j \equiv \begin{cases} \lambda^j \left\{ \lambda \left[(\beta\lambda^2)^{-1-j} - 1 \right] - (1-\delta) \left[(\beta\lambda^2)^{-j} - 1 \right] \right\} \psi & \text{if } j < q \\ - (1-\delta-\lambda) \left[(\beta\lambda^2)^{-q} - 1 \right] \lambda^j \psi & \text{if } j \geq q \end{cases}$$

3. Discussion / Extensions

Recessions as Liquidation Periods (3)



3. Discussion / Extensions

The efficiency of news-driven booms and busts

- ▶ In that basic model, fluctuations are constrained-efficient.
- ▶ But our core mechanism does to rely on constrained efficiency.
- ▶ Assume a constant labor market wedge: $w = \alpha(1 + \tau)$, $\tau > 0$
- ▶ Same positive implications
- ▶ Different normative ones: in a noise driven boom, more employment and consumption $\rightsquigarrow$ the economy gets closer to an efficient outcome.
- ▶ The boom is a good period even if expectations are not met eventually.
- ▶ (This is *not* an unconditional statement)
- ▶ But one can think of a wedge that varies with the business cycle.

3. Discussion / Extensions

The information structure

- ▶ We have a super simple information structure
- ▶ Could be extended with regime shifts for the noise variance + learning about the regime in which the economy is.
- ▶ Information could be gradually revealed instead of in one shot,
- ▶ etc...

3. Discussion / Extensions

Excluding technological regress

- ▶ With Gaussian innovations, technological regress is possible.
- ▶ We have seen that *revisions* are enough to generate busts
- ▶ Therefore, it is possible to generate booms and busts with technological processes that never regress.

3. Discussion / Extensions

Expectations of new markets as a form technological news

- ▶ News need not to be about future productivity.
- ▶ Assume

$$C_t = N_t^\xi \left(\sum_{i=1}^{N_t} X_{it}^\sigma \right)^{\frac{1}{\sigma}}, \quad \sigma < 1$$

- ▶ N_t is now the driving force, and expands over time, with news and noise.
- ▶ Symmetric equilibrium:

$$C_t = N_t^{\xi + \frac{1}{\sigma}} \bar{X}_t$$

- ▶ Hence, much of the literature on technological news can be re-interpreted as models with news of expanding market.
- ▶ Assume $\xi = -\frac{1}{\sigma}$, monopoly power for each intermediate and a setup cost to implement a new variety: totally inefficient business cycles driven by changes in N_t .

Bonus: Why are news shocks challenging for theory?

Intuition (1)

- ▶ Think of an optimal growth model with fixed labor supply.
- ▶ Without shocks to current fundamentals, the economy has to move along the resource constraint

$$C + I = F(K, \bar{L})$$

- ▶ $\rightsquigarrow$ no BC movements are possible: if $C \nearrow$, then $I \searrow$

Bonus: Why are news shocks challenging for theory?

Intuition (2)

- ▶ If labor supply is not fixed, the impossibility result is not changed if leisure is a normal good:

$$C + I = F(K, L)$$

- ▶ if $C \nearrow$, then $L \searrow$ (because leisure goes up as consumption does), and necessarily $I \searrow$
- ▶ Is this impossibility result general?
- ▶ Let us consider some graphical interpretations.

Bonus: Why are news shocks challenging for theory?

Formally (1)

- We introduce the following setup:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, 1 - L_t)$$

$$C_t + I_t = F(K_t, L_t)$$

$$K_{t+1} = I_t + (1 - \delta)K_t$$

Bonus: Why are news shocks challenging for theory?

Formally (2)



$$C + I = F(K, L) \quad (GM)$$

$$U_1(C, 1 - L)F_2(K, L) = U_2(C, 1 - L) \quad (LM)$$

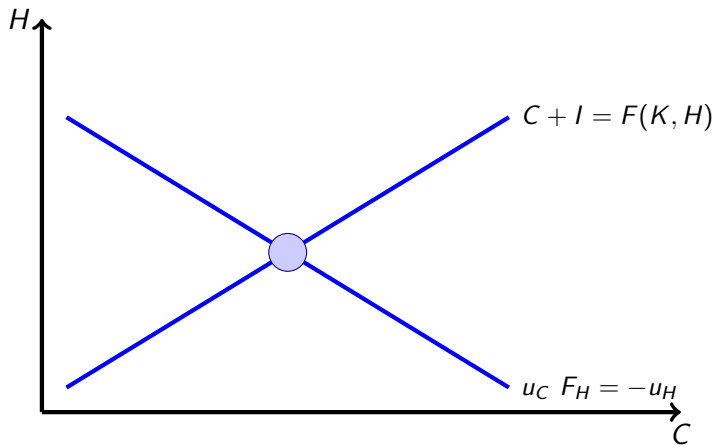
that defines the temporary equilibrium

- ▶ plus a Euler equation

$$U_1(C, 1 - L) = \beta EU_1(C', 1 - L')(1 + F_1(K', L') - \delta)$$

Bonus: Why are news shocks challenging for theory?

A Graphical Interpretation



Bonus: Identification in VARs

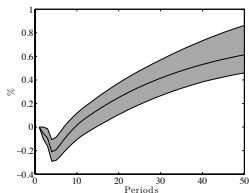
In two slides: identification

- ▶ View the data as being generated by many shocks among which:
 - × Only 2 have a permanent effect on TFP (surprise and news)
 - × Only 1 has a impact effect on TFP (surprise)

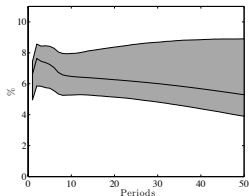
Bonus: Identification in VARs

In two slides: response to a TFP news

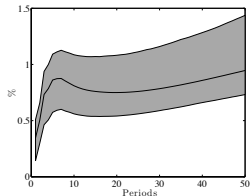
(a) TFP



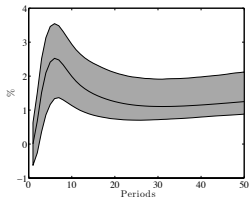
(b) Stock Prices



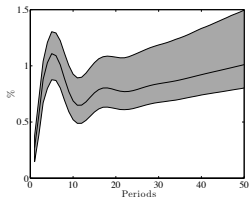
(c) Consumption



(d) Investment



(e) GDP



(f) Hours

